

Design of Sequential Circuits

This example is taken from P. K. Lala, *Practical Digital Logic Design and Testing*, Prentice Hall, 1996, p.176.

Example 1.4 Design a sequential circuit whose state tables are specified in Table 12, using D flip-flops.

Table 12. State table of a sequential circuit.

Present State $Q_0 Q_1$	Next State $x = 0 \quad x = 1$	Output	
		$x = 0$	$x = 1$
0 0	0 0 0 1	0	0
0 1	0 0 1 0	0	0
1 0	1 1 1 0	0	0
1 1	0 0 0 1	0	1

Table 13. Excitation table for a D flip-flop.

Output Transitions $Q \rightarrow Q_{(next)}$	Flip-flop inputs D
0 $\rightarrow$ 0	0
0 $\rightarrow$ 1	1
1 $\rightarrow$ 0	0
1 $\rightarrow$ 1	1

Next step is to derive the excitation table for the design circuit, which is shown in Table 14. The output of the circuit is labeled Z.

Present State $Q_0 Q_1$	Next State $Q_0 Q_1$	Input x	Flip-flop Inputs $D_0 \quad D_1$		Output Z
0 0	0 0	0	0	0	0
0 0	0 1	1	0	1	0
0 1	0 0	0	0	0	0
0 1	1 0	1	1	0	0
1 0	1 1	0	1	1	0
1 0	1 0	1	1	0	0
1 1	0 0	0	0	0	0
1 1	0 1	1	0	1	1

Table 14. Excitation table

Now plot the flip-flop inputs and output functions on the Karnaugh map to derive the Boolean expressions, which is shown in Figure 16.

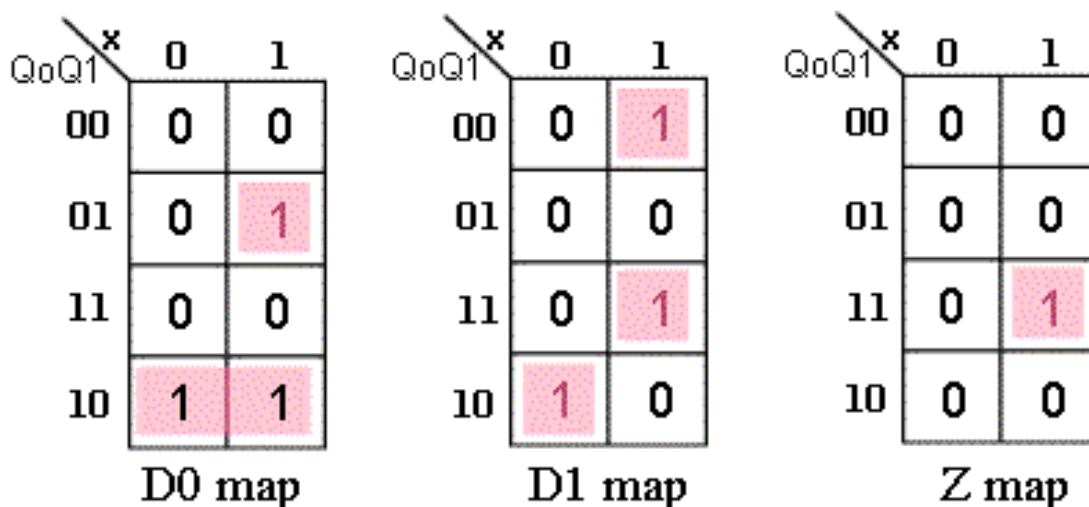


Figure 16. Karnaugh maps

The simplified Boolean expressions are:

$$D_0 = Q_0 * Q_1' + Q_0' * Q_1 * x$$

$$D_1 = Q_0' * Q_1' * x + Q_0 * Q_1 * x + Q_0 * Q_1' * x'$$

$$Z = Q_0 * Q_1 * x$$

Finally, draw the logic diagram.

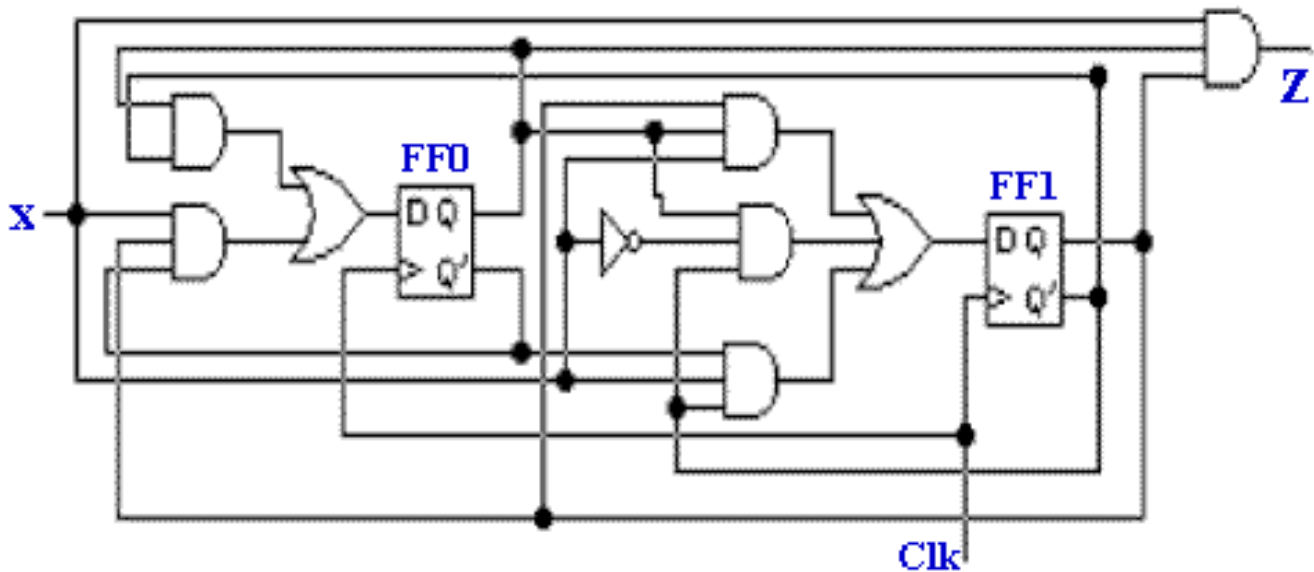


Figure 17. Logic diagram of the sequential circuit.



[Back](#) [Home](#) [Next](#)